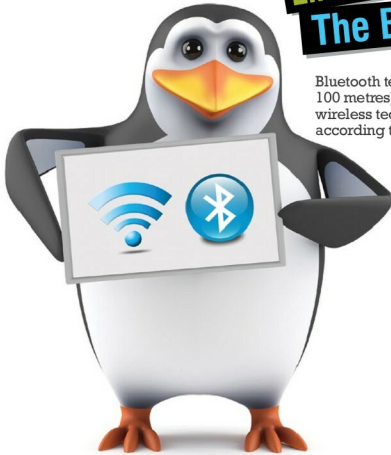


# Linux Without Wires

## The Basics of Bluetooth



Bluetooth technology is used for short range (1 metre to 100 metres) communication. It is the most widespread wireless technology, and is divided into classes according to the power and communication range.



**B**luetooth is a packet based protocol and has a master-slave structure. It works between the 2400MHz and the 2483.5MHz frequency range, and uses the *frequency-hopping spread spectrum*. Bluetooth divides data into packets and transmits them on one of the 79 designated channels. Each channel has a 1MHz bandwidth. Version 4.0 of Bluetooth has 40 channels, and each channel uses a 2MHz bandwidth.

### Bluetooth stack

A Bluetooth stack is software used to implement the Bluetooth host protocol stack. Bluetooth stack implementation can be of two types:

- The first is a general-purpose implementation that is usually for desktop computers, and it is flexible because additional Bluetooth profiles can be added through drivers.
- The second is an embedded system implementation, which is for devices where resources are limited, like peripheral devices.

There are various types of Bluetooth stacks in the market like BlueZ, Bluedroid, Widcomm, Bluecore+, BlueLet, BlueMagic, etc (see Figure 2).

### BlueZ

BlueZ is a general-purpose Bluetooth stack that is used to implement the Bluetooth host protocol stack for Linux. It is open source software and the official Linux Bluetooth stack. You can download this software from <http://www.bluez.org/> in the development section.

BlueZ maps Bluetooth protocol layers to kernel modules, kernel threads, user space daemons, configuration tools, utilities and libraries (see Figure 3). The main components of BlueZ are:

- *bluetooth.ko*, which contains core infrastructure of BlueZ. It exports sockets of the Bluetooth family – *AF\_BLUETOOTH*. All BlueZ modules utilise its services.
- Bluetooth HCI packets are transported over UART or USB. The corresponding BlueZ HCI implementation is *hci\_uart.ko* and *hci\_usb.ko*.
- The L2CAP layer of Bluetooth, which is responsible for segmentation, reassembly and protocol multiplexing, is implemented by *l2cap.ko*.
- With the help of *bnep.ko*, TCP/IP applications can run over Bluetooth. This emulates an Ethernet port over the L2CAP layer. The kernel thread named *kbnpd* is responsible for BNEP connections.